23 For example:



986 → 3 D A

- Keep dividing by 16 until the value 0 is reached.
- Keep track of remainders following each division.
- The hexadecimal number is obtained from the reminders reading from bottom to top.
- It is necessary to convert remainders that are greater than 9 into the hex characters A, B, C, D, E or F.

24 **a** binary digit

b
- (computer) circuits are made up of switches/transistors …
- … which can be ON (1) or OFF (0)

c
- hex is a more human-friendly representation of binary values …
- … since it uses only 2 digits to represent an 8-bit binary number (greater information density)
- it is easy to interconvert between hex and binary …
- … which makes it easier when writing software or diagnosing errors in software

2 Data transmission

1 **a** A data packet is when message/data is split up into smaller groups of bytes:
- for transmission over a network
- containing a header, payload and a trailer

b In any order:
- IP address of the sender
- IP address of the recipient
- the sequence number of the data packet
- the size of the packet (in bytes)

c
- The payload contains the actual data being sent in the packet.
- The trailer is a way of identifying the end of a packet and it also contains an error-checking method, such as a cyclic redundancy check (CRC).

© David Watson and Helen Williams 2021

d i
- The 500 KiB file is split up into a number of packets.
- Each packet will follow its own path from 'A' to 'B'.
- Routers (R1 to R10) will determine the path of each data packet.
- The routing selection depends on the number of packets (at each router).
- The shortest path available is always selected.

ii
- Data packets frequently reach the destination in a different order to the one in which they are sent …
- … which means the data packets need to be re-assembled into the correct order.

2 a Three from:
- There is no need to tie up a single communications line.
- It is possible to overcome failed, busy or faulty lines by re-routing packets.
- It is relatively easy to expand package usage.
- A high data transmission rate is possible.

b In any order:
- Packets can be lost and need to be re-sent.
- The method doesn't work well with real-time streaming.
- There is a delay at the destination while waiting for all of the packets to arrive and then to reassemble them in the correct order.

c The network would grind to a halt as the number of lost packets mount up, 'clogging up' the system.

3 a

Description of transmission taking place	**Simplex (✓)**	**Half-duplex (✓)**	**Full-duplex (✓)**	**Serial (✓)**	**Parallel (✓)**
Data sent one bit at a time down a single wire in one direction only	✓			✓	
Data sent 16 bits at a time down 16 wires in both directions, but not at the same time		✓			✓
Data sent 16 bits at a time down 16 wires in both directions simultaneously			✓		✓
Data sent 8 bits at a time down 8 wires in one direction only	✓				✓
Data sent one bit at a time down a single wire in both directions simultaneously			✓	✓	
Data sent one bit at a time down a single wire in both directions, but not at the same time		✓		✓	

© David Watson and Helen Williams 2021

b

Statements	Serial (✓)	Parallel (✓)
Transmission method used by the memory bus inside a computer		✓
Data can be skewed (out of synch) when travelling over long distances		✓
Least expensive of the two types due to fewer hardware requirements	✓	
Most appropriate where faster transmission rate is essential		✓
Suffers from less risk of external interference	✓	

4 a

- The computer automatically detects a device being connected.
- The device is automatically recognised and appropriate device driver software is loaded up.
- If a new device is connected for the first time, the computer will look for a device driver that matches the device or the user is prompted to download the device driver from the internet.

b

Statements	True (✓)	False (✓)
USBs use a protocol that allows for error-free data transmission between device and computer	✓	
USBs can support any cable length between device and computer		✓
USBs use serial data transfer	✓	
USB connections can transfer data using half-duplex or full-duplex	✓	
It is difficult to add more USB ports to a computer using USB hubs		✓
USB data transfer rates are much faster than, for example, Ethernet connections		✓
There is no need for devices to have a power source because USB cable supplies 5V of power	✓	

© David Watson and Helen Williams 2021

5 **a** pass, fail, fail

b to check for errors following data transmission

c **i** bit number = 4, byte number = 6

ii bit 4 has odd parity (five 1-bits), byte 6 has odd parity (five 1-bits)

iii corrected byte is: 0 1 1 1 1 1 0 1

iv if even parity retained (for example, if two bits have changed)

v
- **checksum**: a method which calculates a check sum from the block of data being sent; the value is sent at the end of the block of data
- **echo check**: a method where the data is sent back to the sender which then checks that it matches the data originally sent
- **Automatic Repeat Request (ARQ):** a method that uses time-out and acknowledgement to automatically request the re-sending of data

6 **a**
- A checksum is calculated from a block of data.
- The calculation is done using an agreed algorithm.
- The checksum is transmitted with the block of data.
- At the receiving end, the checksum is recalculated by the computer using the block of data just sent (using the same agreed algorithm).
- The recalculated checksum is compared to the checksum sent with the block of data.
- If both checksums are the same, then no transmission errors have occurred; otherwise a request to re-send the data block is made.

b
- A copy of the data is sent back to the sender.
- The returned data is compared to the original data sent by the sending computer.
- If there are no differences, then the data has been sent without error.
- If the two sets of data have differences then an error has occurred at some stage during the data transmission.

c
- They use an acknowledgement and time-outs.
- When a receiving device detects an error following data transmission, it asks for the data packet to be re-sent.
- If no errors are detected, a positive acknowledgement is sent to the sender.
- The sending device will re-send the data packet if:
  - a request to re-send data has been received, OR
  - a time-out has occurred
- The whole process is continuous until the data packet received is correct or until the time-out is reached.

7 **a** In any order:
- an incorrect digit entered, for example, 5327 instead of 5307
- a transposition error occurred, for example, 5037 instead of 5307
- omitted or extra digits, for example, 537 or 53107 instead of 5307
- phonetic errors, such as 13 (thirteen) instead of 30 (thirty)

© David Watson and Helen Williams 2021

b i 9 **7** 8 **1** 4 **7** 1 **8** 6 **8** 6 **7**

9 + 8 + 4 + 1 + 6 + 6 = 34

3 × (**7 + 1 + 7 + 8 + 8 + 7**) = 3 × 38 = 114

34 + 114 = 148

148/10 = 14 remainder 8

10 – 8 = 2; the check digit

ii 9 **7** 8 **1** 4 **7** 1 **8** 0 **7** 2 **1** 8

9 + 8 + 4 + 1 + 0 + 2 + 8 = 32

3 × (**7 + 1 + 7 + 8 + 7 + 1**) = 3 × 31 = 93

32 + 93 = 125

125/10 = 12 remainder 5

Therefore, the check digit is incorrect – an input error has occurred.

8 a E

b B

c B

d E

e A

f E

g A

h B

i D

j A

9 a A data packet header doesn't contain MAC addresses. It contains the IP addresses of the sender and recipient.

b Half-duplex refers to data sent in both directions, but not at the same time. The use of four bits sent along four channels refers to parallel transmission.

c Although USB has become an industry standard, this has nothing to do with the orientation of connectors. USB connectors can only be fitted in one way.

d Odd parity is to do with the number of 1-bits and nothing to do with the denary value of the number. Odd parity means a byte contains an odd number of 1-bits.

e Even parity is measured by checking if an even number of 1-bits are sent. A number such as 0 1 1 0 1 1 0 0 1 has four 0-bits but has five 1-bits and is therefore not showing even parity.

f Check digits are used to check if data has been entered correctly. They are not used to check for transmission errors.

g A checksum is calculated from bytes in the data block and not from the 1-bits in a single byte.

h Asymmetric encryption uses two separate keys – a public key and a private key.

© David Watson and Helen Williams 2021